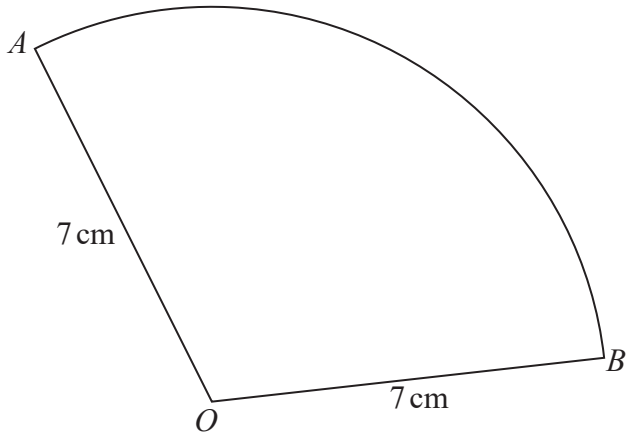
12 *OAB* is a sector of a circle with centre *O* and radius 7 cm.



The area of the sector is 40 cm^2

Calculate the perimeter of the sector.
Give your answer correct to 3 significant figures.

Full circle area = $\pi \times 7^2 = 49\pi$ cm^2

1 mark
Find area of full circle.

Sector is $40/(49\pi)$ of a full circle.

1 mark
Sector fraction $40/(49\pi)$.

Arc = $40/(49\pi) \times 2\pi \times 7 = 80/7$ cm

1 mark
Use fraction for arc length.

Perimeter = $7 + 7 + 80/7$

= 25.42857... cm ≈ 25.4 cm (3 s.f.)

1 mark
Perimeter 25.4 cm.

25.4 ..... cm

(Total for Question 12 is 4 marks)